YEN-LUN (ALLEN) CHOU

(+886) 986-818-060 | 30033allen@gmail.com | Github : Allenchou0708

EDUCATION

National Chengchi University (NCCU) Sep. 2020 – Jun. 2024

- B.E. in Computer Science
- Overall GPA: 4.24/4.30; Rank: 2/60
- Relevant Courses: *Convolutional Neural Networks and Visual Recognition, Natural Language Processing, Video Compression, and Generative AI: Text and Image Synthesis*

RESEARCH INTERESTS

Robot Perception and Learning, Deep Learning, Foundation Model

PUBLICATIONS

- **Yen-Lun Chou**, Pei-Heng Chiang, Kuan-Yan Chu, Yi-Cheng Tseng, Ting-Yun Huang, Kuan-Hung Lin, Hsuan-Tien Lin, Yan-Tsung Peng, and Jun-Cheng Chen (2026) "High-Resolution Regional Weather Adaptation with Parameter-Efficient Fine-Tuning and Retrieved Global Dictionary" *Submitted to IEEE International Conference on Big Data.* (Under Review; First co-author)
- **Yen-Lun Chou**, Pei-Heng Chiang, Yuan-Yu Lim, Yan-Tsung Peng, and Jun-Cheng Chen (2026) "IPQ-PhyHD: Interpretable PQ Dictionary Retrieval with Physical PEFT for HD Regional Weather Adaptation" *Submitted to Association for the Advancement of Artificial Intelligence (AAAI).* (Under Review; First co-author)

RESEARCH EXPERIENCE

Next-Generation Human-Robot Collaborative AI Agents Apr. 2026 - Present
Research Assistant, AIIU Lab (Supervisor: Dr. Jun-Cheng Chen), Academia Sinica (CITI)

- Advanced to Phase II of the EBiM Competition, developing a deep learning-based robotic system for picking, transporting, aligning, and placing thermal pads onto PCBs. Fine-tuned the $\pi_{0.5}$ Vision-Language-Action (VLA) model and applied GRPO reinforcement learning for robotic manipulation.
- Built and configured a UMI and SO101 robotic platform for robot teleoperation and demonstration data collection.

Generative AI Technologies for Addressing Localized Needs Nov. 2025 - Present
Research Assistant, AIIU Lab (Supervisor: Dr. Jun-Cheng Chen), Academia Sinica (CITI)

- Developed a lightweight dictionary learning module for regional weather forecasting to recover missing global contextual information, achieving improved efficiency compared with the YingLong model.
- Incorporated physics-informed constraints at multiple processing resolutions, improving model interpretability and achieving a 4% reduction in RMSE.

AWARDS & SCHOLARSHIPS

- Academic Excellence Award, NCCU (Spring 2021, Spring 2022): Awarded to top 5% of students each semester